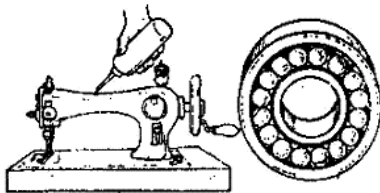


Solution

FRICTION -03

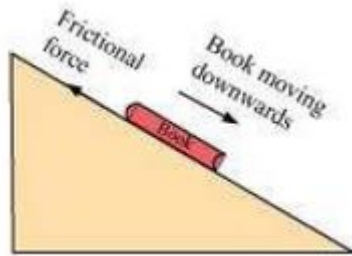
Class 08 - Science

1. Interlocking of irregularities is avoided.
2. Commonly gases and liquids are called fluids.
3. Boats, aeroplanes, all automobiles and ships.
4. We would not be able to walk on the surface if there is no friction on the floor. It is the friction which helps us to stand or walk on the surface. The grooves of our feet or shoes are locked into the irregularities of the floor and make us stand.
5. Ridges of our palm help us to make firm grip. When we hold something by our hands. Ridges of the bottom of our feet help us to when walking and we do not slip because of these.
6. Use of ball bearing will reduce the friction.
7. Air is very light and thin. Yet it exerts frictional force on objects moving through it. Similarly, water and other liquids exerts force of friction when objects move through them. In science, the common name of gases and liquids is fluids. So we can say that fluids exert force of friction on objects in motion through them.
The frictional force exerted by fluids is also called drag.
8. Frictional force responsible for wearing out of bicycle tyres..
9. Air cushion is used to reduce friction in machines where lubrication are not advisable.
10. No, we cannot eliminate friction completely.
11. Ceiling fan.
12. Friction depends on the speed with respect to the fluid and the nature of fluid.
13.
 - i. Spike increases friction by roughening the surface of the shoes.
 - ii. This increase ground grip and decrease your risks of slipping.
 - iii. It becomes easier for athletes to walk or run when their grip on the ground improves.
14. When luggages are fitted with rollers, the resistance to motion is decreased by offering a smooth flow of motion. Hence, friction is reduced and rolling of luggage is anyday easier than lifting them.
15. When a body rolls on other body then rolling friction comes into play.
16. A shape which is wide in middle and tapered at ends is called a streamlined shape. Such a shape reduces drag or fluid friction.
17. Friction is caused by the irregularities on the two surfaces in contact. The irregularities on the two surfaces lock into one another.
Due to the interlocking of two surfaces friction arises.
18. Ball bearing change the sliding friction into rolling friction, as we know rolling friction is less than sliding friction and hence the friction is reduced.
19. Talcum powder is used in carrom board to reduce friction.
20. Brake pads.
21. Friction will be greater in case of a rough surface.
22. When an object moves through a fluid, the fluids create friction. Friction created by a fluid is called fluid friction or drag.
23. Brake pads is used in cycle brakes to increase friction.
24.
 - i. by applying a lubricant.
 - ii. by using ball bearing.
 - iii. by polishing surfaces
 - iv. In the machines where oil cannot be used air cushion between the parts is used to reduce friction.



25. When a book slides on the writing desk, a frictional force acts between the book and the surface of the desk. The direction of frictional force on the book is opposite to the direction of its motion and acts in upward direction, as shown in the following

figure.



26. Friction can also produce heat. When we rub our hands together vigorously for few minutes we feel hot. When we strike a match stick, against the rough surface, it catches fire. We also observe that the jar of a mixer becomes hot when it is run for a few minutes.
27. No, the box will not experience any force of friction because the force applied by both the boys is equal. Thus, the net (total) force on the box will be zero. As a result, there will be zero force of friction.
28. No, he will need the same effort because even by placing it vertically there will be same amount of friction as friction does not depend upon the area of contact.
29. Yes, the force of friction comes into play when a raindrop rolls down a glass window pane. The force of friction always takes place when two surface comes in contact. Here one is the raindrop's surface and the other is the glass window pane's surface.
30. The block of 2 kg iron will exert more friction. As we know, friction increases with the increases in weight.
31. (i) We sprinkled fine powder on the carromboard.
(ii) When a few drops of oil are poured on the hinges of a door, the door moves smoothly
(iii) A cycle or a motor mechanic uses grease between the moving parts of the cycle.
- In all the above cases, we want to reduce friction in order to increase efficiency. When oil, grease or graphite is applied between the moving part of a machine, its thin layer is formed there and moving surfaces do not directly rub against each other. Interlocking of irregularities is avoided to a great extent. Movement becomes smooth. The substances which reduce friction are called lubricants. In some machines it may not be advisable to use oil as lubricant. There is an air cushion between the moving parts is used to reduce friction.
32. There are two factors on which friction depends:
- Friction depends on the nature (smoothness or roughness) of surfaces in contact.
 - Friction depends on how hard the two surfaces press together.
33. (i) Friction depends on the nature (smoothness) of surfaces in contact.
(ii) Friction depends on how hard the two surface press together.
34. When objects move through fluids, they have to overcome the friction acting on them. Efforts are therefore made to minimize the friction so objects are given special shapes. Idea of such shapes come from the body structure of birds and fishes which have to move about in fluids all the time. Such shapes are called streamlined. Giving such shapes to the ships, plane and cars is called streamlining.
35. When used for a long time, slippers with rubber soles become slippery because, for a given pair of surfaces, friction depends upon the smoothness of those surfaces. Due to the continuous rubbing of slippers with the ground, the rubber soles(surface) of slippers become smooth. As a result, friction decreases between the rubber sole and the ground. Thus, slippers become slippery.
- 36.
- We are able to walk because of the friction present between our feet and the ground.
- In order to walk, we push the ground in the backward direction with our feet. The force of friction pushes it in the forward direction and allows us to walk. The force of friction between the ground and feet decreases when there is soapy water spilled on the floor. Hence, it becomes difficult to walk on the soapy floor.
37. (i) We deliberately increase friction by using brake pads in the brake system of bicycles and automobiles.
(ii) When we are riding a bicycle, the brake pads do not touch the wheels. But when we press the brake lever, these pads arrest the motion of the rim due to friction. The wheel stops moving.
(iii) Kabaddi players rub their hands with the soil for an easy grab of their opponents.
(iv) Gymnasts apply some coarse substance on their hands to increase friction for better grip.
38. Spring balance is a device used for measuring the force acting on an object. It contains a coiled spring. The spring gets stretched when a force is applied on it. Stretching of the spring is measured by a pointer moving on a graduated scale. The reading on the scale gives the magnitude of force.



Spring balance.

39. Examples of Smooth Surfaces:

- (i) Surface of the mirror
- (ii) Floor of the room when polished,
- (iii) Surface of wet soaps,
- (iv) Glazed tiles,
- (v) Surface of an oily pot.

Examples of Rough Surfaces:

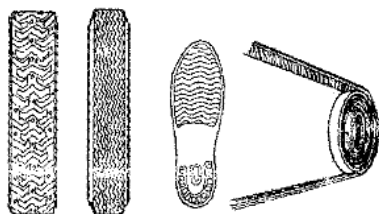
- (i) Soles of shoes,
- (ii) Surface of ply board
- (iii) Unpainted wall
- (iv) Surface of a brick,
- (v) Surface of tyres.

40. The objects are given special friction. Aeroplane and birds both fly in the air and have to face friction exerted by air. They are so shaped that they do not have to lose energy while overcoming the frictional force exerted by air to fly. Therefore the aeroplanes are also shaped streamlined, so that they can overcome the frictional force of air.



41. When one body rolls over the surface of another body, the resistance to its motion is called the rolling friction. Rolling reduces friction. It is always easier to roll than to slide a body over another. That is the reason it is convenient to pull the luggages fitted with rollers.
42. When a spacecraft re-enters the atmosphere, its surface gets extremely hot because of the friction between the air and the spacecraft moving at a very high speed. To protect the craft its outer surface is covered with tiles of a special material, which can tolerate very high heat. The reason for the burning of spacecraft Columbia in which Kalpana Chawla was travelling was the heat by friction.
43. Due to the interlocking of the irregularities on the two surfaces in contact, the force of friction arises. On the floor when a heavy object is placed, the interlocking of irregularities on the surface of box and floor become strong. This is because the two surfaces in contact are pressed harder. Hence, more force is required to overcome the interlocking. Thus, to push the heavier box, Seema has to apply greater force than Iqbal.
44. Friction is an evil. It wears out the materials whether they are screws, ball bearings or soles of shoes. Worn out steps of foot overbridges on railway stations are also due to friction. Friction can also produce heat. Friction produces heat in moving parts of machine which results in wastage of energy and fuel. Friction causes wear and tear of moving parts. The shoe sole gets flattened after few months due to the friction offered by the ground.
45. The friction force on an object in a fluid depends on two factors:
- a. Speed of the object in fluid.
 - b. Shape of the object and nature of the fluid.
46. To minimize the friction due to water or any other liquid and air or any gas, the moving body should have a streamlined shape. The shape of the body which offers least resistance due to friction is called streamlined shape.

47. It is done to provide the shoes better grip on the floor, so that one can move safely. Similarly, the treaded tyres of cars, trucks and bulldozers provide better grip with the ground.



48. Since the rolling friction is smaller than the sliding friction, sliding is replaced in most machines by rolling by the use of ball bearings. A common examples are the use of ball bearing between the hub and the axle of a ceiling fan and the bicycle.
49. The marble will move the shortest distance on the layer of sand because the silk cloth and glass sheet will exert lesser maximum friction whereas the sand (being more rough) exert maximum friction against the motion.
50. The friction is due to the interlocking of irregularities in the two surface. When the object starts sliding the contact points on its surface, do not get enough time to lock into the contact points on the floor. So the sliding friction is slightly less than the static friction.
51. (i) The frictional force on an object in a fluid depends on its speed with respect to the fluid.
(ii) The frictional force also depends on the shape of the object and the nature of the fluid.
52. It is obvious that when objects move through fluids, they have to overcome the friction acting on them. In this process they lose energy. Efforts are, therefore, made to minimize the friction. So objects are given special shapes. The idea of special shapes comes from nature. Birds and fishes have to move about in fluids all the time. Their bodies must have evolved to shapes which would make them not to lose much energy in overcoming friction. Such shapes are called streamlined. This is the reason ships, aeroplanes and cases are given a shape which resembles the streamlined shapes of birds and fishes.
53. While playing tug of war, Preeti felt that the rope was slipping through her hands. To prevent this, first, she needs to make her smooth hand non-smooth. Because the friction decreases as the surface becomes smooth. Hence, the rope was slipping through her hands (smooth). To make her hands non-smooth, she should apply sand on both hands. Due to the sand, her hands become non-smooth (rough). Friction increases when the surface is rough. As a result, friction increases between the rope and her hand, and now the rope won't slip through her hands.

54. Factors affecting Friction

1. **Nature of the Surface-** The force of friction depends on the nature of the surfaces in contact i.e. smooth surface or rough surface. As we know, friction is caused by irregularities on the surfaces in contact so if the surfaces are smooth then the number of irregularities will be less hence the friction will also be less, but if the surfaces in contact are rough then they have more irregularities on the surface. Such surfaces when slides over each other, gets interlocked due to their irregularities. More the resistance to motion, more will be the friction. So larger force is required to make them move over each other.
 2. **Mass of the Body-** The force of friction depends on the mass of the body which is lying on the surface. More the mass of an object, more will be the friction. For example - One book is lying on the table, a gentle push to the book can it move. But if we put two more of such books having same thickness, more force is required to overcome the friction offered by the table to the books. Hence it is clear that more the mass of the object, more will be the friction.
55. **Static friction** is friction between two or more solid objects that are not moving relative to each other. For example, static friction can prevent an object from sliding down a sloped surface.
- The friction between the two surfaces of the bodies which are in contact with each other and also at sliding relatively is called **sliding friction**.

56. (i) Friction as a Friend:

Frictional force is very useful in several situations. If there were no friction between the wheels and the road, no vehicle would be able to move. If there were no brakes in a bicycle there would be many accidents. It is due to the frictional force between the road and the wheel or between the wheel and the brake that vehicles can move and stop.

In the same way, when we walk, because of friction the sole of our footwear rubs against the hard surface below. If there were no friction we would slip. The reason for slipping on a banana peel is because the friction is low.

When two objects are rubbed against each other heat is produced. That is the reason you rub your hands together in winter. But for friction it would not be possible to light a matchstick.

(ii) Friction as a Foe:

Machine parts that come in contact with one another wear out because of friction. Similarly the soles of our shoes wear out because of friction.

57. Yes, rolling force works between the wheels and iron nails. The friction between these will decrease if a cushion will be introduced between them.
58. Friction is harmful in many ways.
- It wears out the material whether they are screws, ball bearing or soles of shoes.
 - Worn out steps of foot over bridges on railway station are also due to friction.
 - Friction can also produce heat.
 - Friction causes moving objects to stop or slow down.
59. The surface of mortar and pestle used for grinding is etched again after prolonged use because the mortar and pestle lose their roughness after prolonged use (using for long time). As a result, the force of friction decreases and they do not grind properly. So they are etched to make them more effective for grinding again.
60. The handle of a cricket bat or badminton racquet is usually rough because we know that the force of friction increases as the surface is rough. The surface of the bat or badminton racquet is rough, therefore it does not slip away through the hands. As a result, there will be a better grip between the handle of the bat and hands due to increased friction.
61. The one rubbed with sandpaper shines more than the other because when the metal sheet is rubbed with sandpaper, it produces more friction between the metal sheets and sandpaper. As a result, the force is enough to remove easily all the dust particles present on the metal sheet, so it shines more. On the other hand, when the metal sheet is rubbed with ordinary paper, it does not produce enough force of friction to remove dust particles. Thus, the metal sheet rubbed with ordinary paper shines less.
62. We cannot write with a chalk, pen, pencil if there were no friction. If we are writing with a chalk on frictionless surface then no chalk particles stick to the surface. So we cannot write with a chalk if there were no friction. On the other hand when we are writing with a chalk on the blackboard, its rough surface rubs off some chalk particles which stick to the blackboard.
63. **Advantages of friction:**
- (i) We cannot write with a pen or pencil, if there were no friction.
 - (ii) We cannot write on the blackboard with a chalk, if there were no friction.
 - (iii) If an object started moving, it would never stop, if there were no friction.
 - (iv) We cannot walk on the road without friction.
 - (v) We cannot fix a nail on the wall or tie a knot without friction.
- Disadvantages:**
- (i) It wears out the materials whether they are screws, ball bearings or soles of shoes.
 - (ii) It causes damage to the parts of machines.
 - (iii) Friction also produces heat. When a machine is operated, heat is generated that causes wastage of energy.
 - (iv) The tyres of cars, buses and trucks, etc., also worn out due to friction.
 - (v) Friction reduces the speed due to which more force is required.
- In this way, we see that friction is harmful as well as useful so it is called necessary evil.
64. If the seat cover is very smooth, we tend to slip when brakes are applied suddenly because as we know that the friction decreases when the surface is smooth. The seat cover of a rickshaw is smooth, this means the friction between the body and the seat is lesser. Thus, when the brakes are applied suddenly, we tend to slip due to decreased friction.
65. The soles of shoes are worn out due to the effect of friction. The soles of shoes increase the friction between surfaces because it makes the surface of the shoes rough. Thus, this is due to the fact that friction causes wear and tear of objects and reduces the life of the object.
66. **Advantages:**
- a. Friction with air makes the windmills work, which are a source of wind energy. Similarly, friction with water makes the turbines work in the hydroelectric power plants.
 - b. Friction of propeller blades with air, propels a glider or a small aircraft in air.
- Disadvantage**
- a. Speed of the object reduces, as fluid friction opposes the motion of the object.
 - b. Energy of the object is lost in overcoming fluid friction when moving through the fluid. This reduces their efficiency.
67. Tyres are made by corrugated in order to increase the friction between the tyres and the surface of the road. This increases the opposing force incurred by the tyres. The rougher the tyre is the grip between the road and the tyre is better. This consequently increases the friction which tends to smooth vehicular motion. Hence prevents accidents due to slipping and sliding.
68. **Rolling friction** is the force resisting the motion when a body (such as a ball, tire, or wheel) rolls on a surface. **Sliding friction** is the force resisting the motion when a body slides on a surface. The force of friction depends on the area of contact between the two surfaces. As the area of contact is less in the case of rolling than in the case of sliding, rolling friction is less than the sliding friction. That is why suitcases and automobiles are fitted with wheels so that it is easier to move them.

69. a. A small quantity of talcum powder on a wooden surface of carom board reduces friction between sticker and carom board surface. The carom coins slide easily over the board, which is desired.
b. If we try to hold a glass with soapy hands, it can easily slip from our hands due to no friction. Glass has smooth surface and after putting soap on hands, they also gets smooth. Friction does not work on smooth surfaces due to no irregularities between glass and soapy hands.
70. Take a ball and allow it to roll on the cemented surface. Measure the distance covered on the surface by the measuring tape. Repeat this activity on a kachha floor also with same force and measure the distance covered by the ball. You observe that ball covers a larger distance on the cemented floor because it is smoother and therefore frictional force between the cemented floor and the ball is less.
We can conclude that friction depends upon roughness or smoothness of the surfaces in contact. Smooth surfaces offer lesser friction than rough surfaces.
71. Place wooden block on the smooth surface. Fix spring balance to hook and try to slide wooden block by pulling spring balance. Read the observation shown by the spring balance.
Now place the 50g of weight on the wooden block and repeat the same activity. Record the force of friction. Repeat this activity with different weights and record the observations.
It can be easily understood that as the weight of wooden block increases, the magnitude of frictional force increases.
72. Shikha has to apply larger force because in case of heavier box the opposing friction force is more than that of lighter box. Friction force is directly proportional to the weight of the object which moves. Hence, larger the weight, greater the pressure between two surfaces. If the two surfaces are pressed harder then there is stronger interlocking, due to which friction is greater. To overcome this friction, one has to apply more force which is greater than friction force.
73. a. **By polishing the surface:** rough surface can be made smooth by polishing it. That is how, friction can be reduce. Polishing removes hills and valleys from the surfaces. If the surfaces, that are in contact are rough there is more friction. After polishing, surfaces become smooth, so it has less friction.
b. **By streamlining the bodies:** to reduce fluid friction, aeroplanes, rockets, sail boats are shaped into streamlined body. Birds have streamed lined bodies.
74. It sometimes desirable to increase friction to avoid slipping. There are two methods to increase friction:
a. By making the surface rough
b. by increasing the mass of the object, that is moving.
For example, by using brakes pads in the system of bicycles and automobiles. When we are riding a bicycle, the brakes pads do not touch the wheels. But when we press the brake lever, these pads arrest the motion of the ring due to friction. Spikes are provided in soles of shoes used by football players and athletes to increase friction so that they get a firm grip on the ground.
75. The kabaddi players rub their hands with the soil before the game start because it will increase friction between their hands and the opponent's body so that they can get a better grip on them. They do so because they don't want their opponents to slip of their hands. Kabaddi players have oil massage so they often get slipped off. But this is prevented due to soil.